

# Hybrid VARMAX-ML Long Short Market Neutral Trading Strategies

Team Telecom - Shaad Hafeez  
Indian Institute of Science Education and Research  
ADVANCED QUANTITATIVE ANALYTICS

May 10, 2026

## Abstract

This report details a daily long-short market-neutral trading strategy. The strategy utilizes Vector Autoregression with Exogenous Variables (VARX) as a baseline model to capture linear trends, combined with Machine Learning algorithms (LSTM, MLP, Gradient Boosting, LightGBM) to capture non-linear patterns missed by the baseline. These predictions are converted into signals and backtested against the S&P 500. Results indicate that the hybrid strategy delivers superior drawdown-adjusted returns and uncorrelated alpha, suggesting it is well-suited for leveraged portfolios and tail-risk protection.

**Keywords:** Vector Autoregression, Dimensionality Reduction, Support Vector Regression, LSTM, Quantitative Finance, Time Series Forecasting.

---

## 1 Introduction

Predicting daily stock returns for several correlated assets, in the case of this study, the universe of American telecommunication stocks, is a difficult task due to the highly noisy nature of returns. Simple linear models often fail at this task because they make an assumption about the underlying distribution of data. Financial time series prices are non-stationary in nature. We could differ the series to remove any non-stationarity, but returns are highly noisy, and it is difficult for any linear model to converge to a solution. This is discussed in detail later. In this study I use a hybrid framework where the Linear model, in this case, is a Vector Auto Regression with Exogenous Variables (VARX), and the combine it with a non-linear machine learning algorithm to see if this hybrid framework delivers superior results than the base linear model.

## 2 The Theoretical Framework

Vector Autoregression is a statistical model that can be used to analyze the relationship between multiple time series at once, as opposed to a simple Autoregressive model. The general equation for a VAR with Exogenous variables (VARX) is :

$$Y_t = c + \sum_{i=1}^p A_i Y_{t-i} + \sum_{j=0}^q B_j X_{t-j} + \epsilon_t \quad (1)$$

where,

- $Y_t$  = Endogenous variables at time  $t$ .
- $C$  = is the intercept vector.
- $p, q$  = are the number of endogenous and exogenous lags.
- $A_t$  = matrix of coefficients of lagged  $Y_t$ .
- $X_{t-j}$  = Exogenous variables at lag  $j$ .

- $B_j$  = Coefficients of lagged  $X_{t-j}$
- $\epsilon_t$  = Error term.

Here  $\epsilon_t$  contains all the non-linear patterns and noise that VARX failed to capture. We can rewrite this equation as

$$Y_t = L_t + \epsilon_t \quad (2)$$

where  $L_t$  is the linear part of the data that can be captured by VAR. Now we can run an ML algorithm to further capture any non-linear patterns that may have remained

$$\epsilon_t = Y_t - \hat{L}_t \quad (3)$$

Where  $\hat{L}_t$  is the linear part captured by VARX. The ML predictor of  $\epsilon_t$ , call it  $\hat{N}_t$

$$\hat{N}_t = f(\mathbf{Features}(\hat{L}_t, Y_{t-1})) \quad (4)$$

where  $f$  is a functional approximation made by the ML algorithm. We have our final predictor which is made of  $\hat{L}_t$  and  $\hat{N}_t = f(\mathbf{Features}(\hat{L}_t, Y_{t-1}))$ :

$$\hat{Y}_t = \hat{L}_t + \hat{N}_t \quad (5)$$

We can say that the ML algorithm acts as a complex Moving Average(MA) predictor of the model. Therefore we can describe our whole model as a hybrid VARMAX-ML model.

### 3 Data

Data is obtained from Yahoo Finance's historical database. 10 years of data is used from January 2016 to 2026. The data used in this study is:

- Daily Closing prices of 10 Telecommunication Stocks. The stocks used are (NASDAQ tickers given) :{**T, VZ, AMX, CHT, TEF, TU, BCE, RCI, VIV, TLK**}.
- Daily S&P 500 index

Both these datasets are from a period of 11/January/2016 to 8/January/2026. I use the price data in the form of a matrix where  $N = 10$  is the number of rows and assets, and  $D = 2514$  is the number of columns or days. Denote the price matrix by  $P_{N,D}$ . Similarly, denote the S&P daily index matrix by  $S_{1,D}$ .

## 4 The Experimental Setup and Design Choices

### 4.1 The Linear Model

The VAR model discussed above is used with S&P as exogenous variable. I first take the natural logarithm of price matrix  $P$ :

$$P_{ij} \rightarrow \log(P_{ij}) \quad (6)$$

Then I apply a 5-day Exponential smoothing to it:

$$\log(P_{ij}) \rightarrow EMA(\log(P_{ij})) \quad (7)$$

I then perform Principal Component Analysis(PCA) to extract 4 most important components from the price data. I train VAR on this extracted components, with s&P as exogenous variable. In parallel I train an ARIMA model on S&P, and get future forecasts which I use to get the test predictions of VAR since it needs future forecasts of S&P to generate predictions. I subtract the predictions from the original raw price data to get residuals. I then proceed to correct the errors made by this VAR model with exogenous variables.

## 4.2 The ML models

The following ML models were implemented to model the residuals of the VARMAX baseline:

- **Multi-Layer Perceptron (MLP):** A simple feed-forward neural network utilized to capture baseline non-linear relationships. Its fully connected architecture provides a flexible approach to mapping input features to the residual targets, acting as a general-purpose function approximator.
- **Long Short-Term Memory (LSTM):** An advanced form of a Recurrent Neural Network (RNN) explicitly designed to overcome the vanishing gradient problem. By utilizing internal gating mechanisms (forget, input, and output gates), the LSTM is well-suited to capture long-term temporal dependencies and sequential volatility clusters within the time series. I use 5 days for time steps.
- **Support Vector Regression (SVR):** A margin-based regression technique that maps input features into a high-dimensional space using a kernel function (such as RBF). SVR aims to fit the prediction error within a specific threshold (an  $\epsilon$ -tube).
- **Gradient Boosting Regressor (GBR):** An ensemble learning method that builds decision trees sequentially. Each successive tree is trained to correct the pseudo-residuals of the previous combined ensemble, creating a robust model that is highly effective at isolating complex, non-linear market signals.
- **LightGBM (LGBM):** A highly optimized, histogram-based gradient boosting framework. Unlike traditional GBR which grows trees level-wise, LightGBM grows trees leaf-wise. This approach allows for faster training speeds, lower memory usage, and often superior accuracy when parsing dense, noisy financial datasets.

## 4.3 ML training and Final Predictions

The following features are generated for the ML models. Using the original raw prices, I generate these three features.

The following features are generated for the ML models:

### 1. Simple Moving Average (5-day)

$$SMA_{ki}(5) = \sum_{j=i-5}^{i-1} P_{kj}/5 \quad (8)$$

where  $k$  denotes the  $k$ th stock.

### 2. Volatility (5-day)

$$VOL_{kj}(5) = \sqrt{\sum_{j=i-5}^{i-1} \frac{(P_{kj} - \sum_{l=i-5}^{i-1} \frac{P_{kl}}{5})^2}{5}} \quad (9)$$

### 3. Rolling Z-score (5-day)

$$Z - SCORE_{ki}(5) = \frac{P_{ki} - SMA_{ki}(5)}{VOL_{ki}(5)} \quad (10)$$

### 4. VAR Prediction

$$\hat{L}_{ki} \quad (11)$$

With these features, the target is:

$$\epsilon_{ki} = P_{ki} - \hat{L}_{ki} \quad (12)$$

Finally the ML model is tasked with predicting the output such that the input is the set of features:

$$\begin{aligned} &\textbf{Features} \longrightarrow \textbf{Target} \\ &(L_{ki}, SMA_{ki}(5), VOL_{ki}(5), Z-SCORE_{ki}(5)) \longrightarrow \epsilon_{k(i+1)} \end{aligned}$$

#### 4.4 ML training and Walk-Forward Validation

ML training is done using the mentioned features and targets. Because financial time series possess strict temporal dependencies, standard  $k$ -fold cross-validation is structurally invalid. A 6-fold walk-forward validation was constructed to simulate true production forecasting, ensuring the models were tested purely on unseen out-of-sample data. An 85 : 15 training-validation split is created. We use the  $R^2$  score to pick the best combination of parameters. We obtain predictions of  $\epsilon_{ij}$  as  $\hat{N}_{ij}$

#### 4.5 Recombination

Finally we recombine to get final predictions of  $P_{ij}$ , call it  $\hat{Y}_{ij}$

$$\hat{Y}_{ij} = \hat{L}_{ij} + \hat{N}_{ij} \quad (13)$$

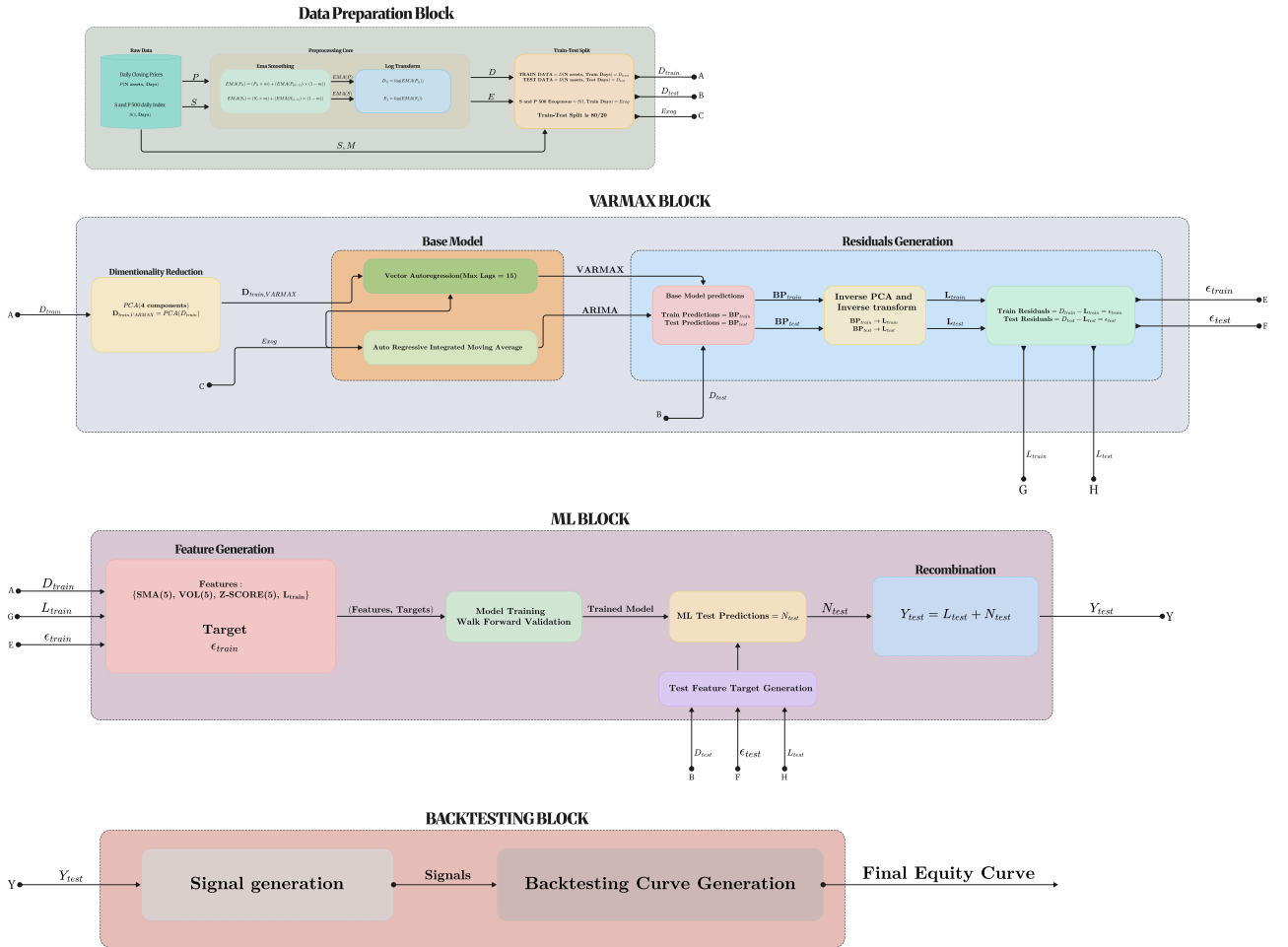


Figure 1: Model Architecture and Pipeline Diagram

#### 4.6 Signal Generation and Backtesting

Signals are generated from the predicted prices to predict next day returns. These signals were smoothed using exponential smoothing in specific cases to reduce turnover costs. The backtest involved selecting the top 3 and bottom 3 assets based on predicted returns (with a 0.1% threshold), applying equal allocation for long and short positions respectively. I denote the resulting backtesting equity curve by  $C$ .

### 5 Testing Statistical Significance of Backtest results

I used two variants of the Monte Carlo permutation test to check statistical significance of the backtest metrics.

## 5.1 Permutation Test against Random Signals

I shuffle the signals generated by the model using a uniformly random permutation of the signals and generate a random signal matrix. I run 5000 iterations of this test and calculate the metrics of the backtest curve generated by this random signal matrix. I compute the p-value of the metric in the following way. Let  $R_i$  denote the random backtest curve and  $T$  denote the true backtest curve. Define

$$I_i = \begin{cases} 1, & \text{if } \text{metric}(R_i) - \text{metric}(T) \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (14)$$

Then we define p-value as

$$\text{p-value} = \frac{\sum_{i=1}^{5000} I_i}{5000} \quad (15)$$

## 5.2 Permutation test against S and P 500

In this case I first compute the difference between the metrics of  $T$  and S&P500. Then I create two fake backtest strategies by uniformly randomly swapping the daily returns of S&P500 and my strategy, compute the difference between their metrics and calculate the p-value similarly.

## 6 Results

Fig 2, 3, 4, 5, 6, 7 show performance of the Models on out of sample data from January 2024 to January 2026. The Blue line is the true prices, and the Red line is the predicted prices

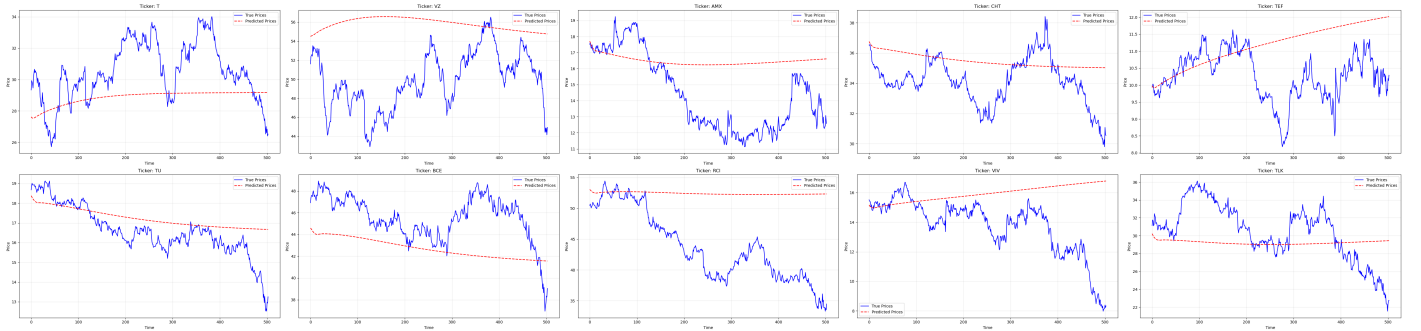


Figure 2: VARX

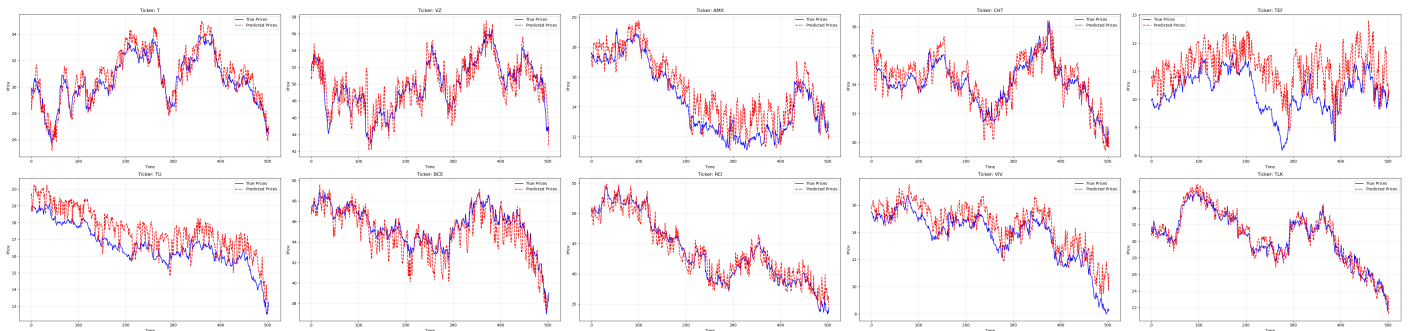


Figure 3: VARMAX-SVR

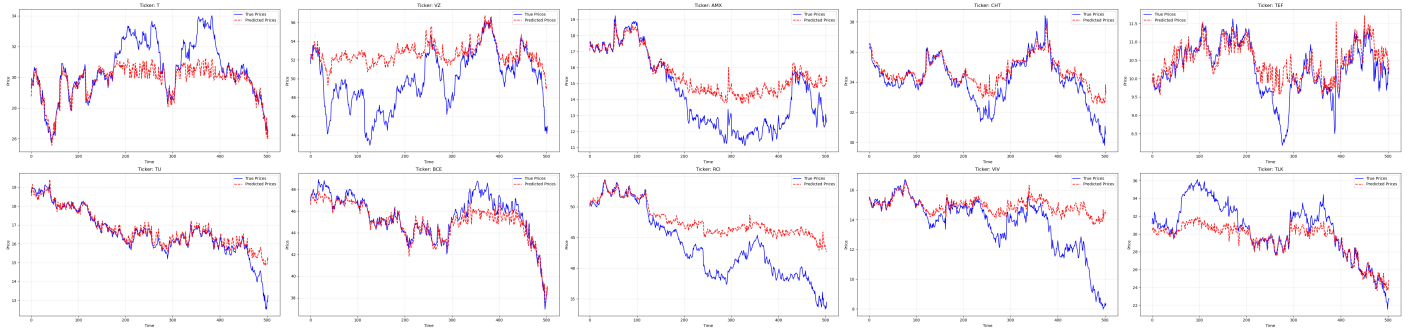


Figure 4: VARMAX-GBR

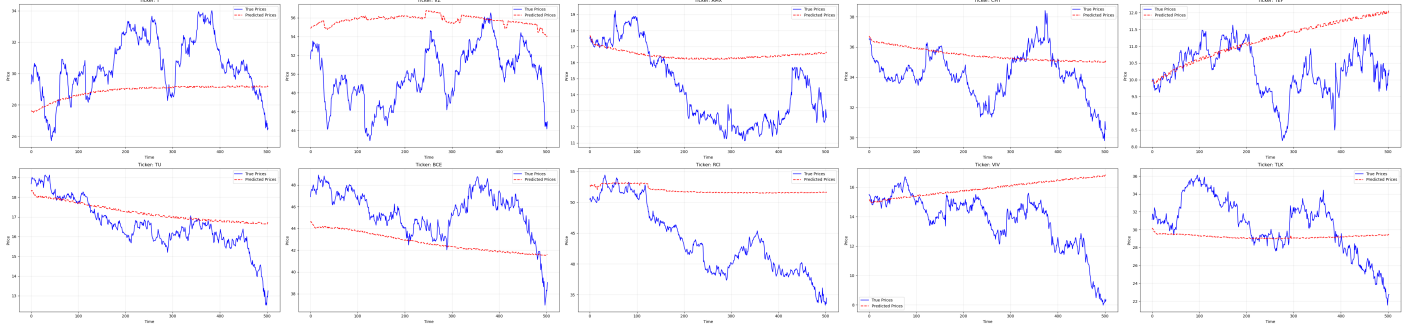


Figure 5: VARMAX-LGBM

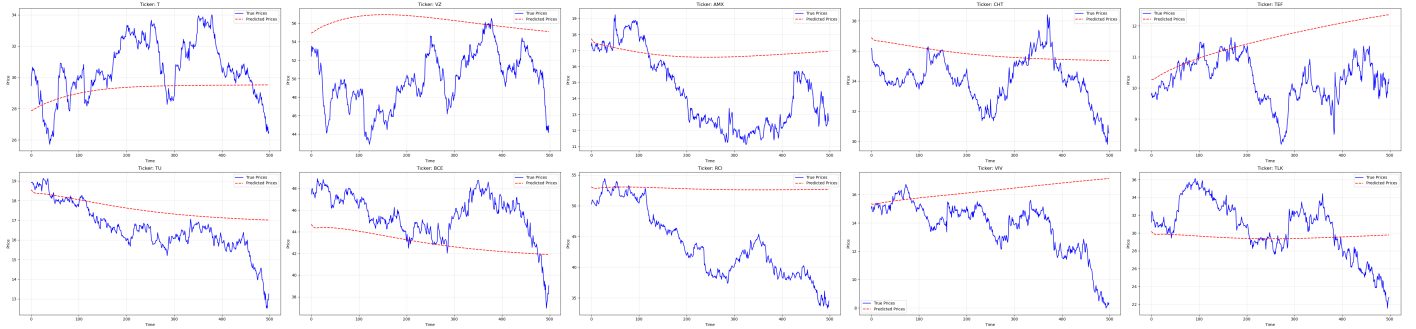


Figure 6: VARMAX-LSTM

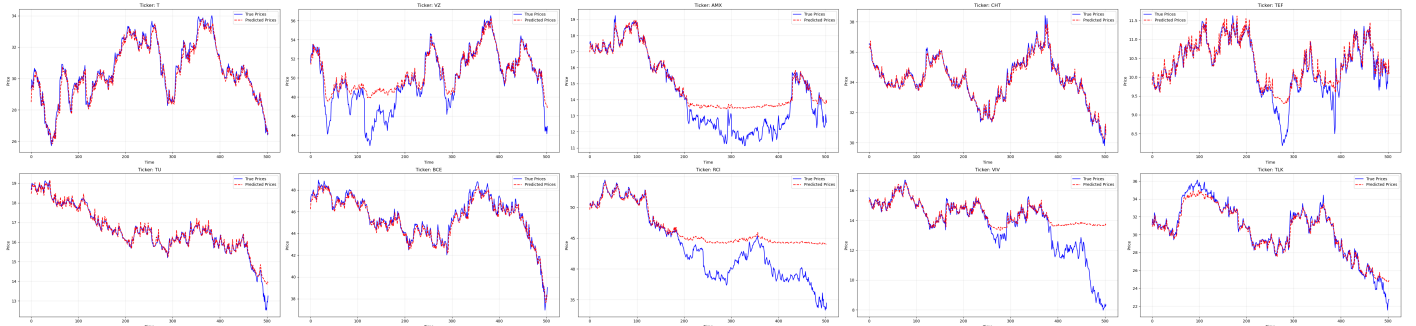


Figure 7: VARMAX-MLP

## 6.1 Model Metrics

The two metrics under discussion here are:

- $R^2$  : tells us how much of the variance in the data is explained by the model.

$$R^2 = 1 - \frac{\sum_{j=1}^N \sum_i^D (Y_{ji} - \hat{Y}_{ji})^2}{\sum_{j=1}^N \sum_i^D (Y_{ji} - \bar{Y}_{ji})^2} \quad (16)$$

- Hit Rate : tells us the directional accuracy of returns, the rate at which the model correctly predicts positive or negative direction

$$\text{Hit Rate} = \frac{\sum_{j=1}^N \sum_{i=2}^D \begin{cases} 1, & \text{if } \text{sgn}\left(\frac{Y_{ji} - Y_{j,i-1}}{Y_{j,i-1}}\right) = \text{sgn}\left(\frac{T_{ji} - T_{j,i-1}}{T_{j,i-1}}\right) \\ 0, & \text{otherwise} \end{cases}}{(N - 1) \times (D - 1)} \quad (17)$$

Table 1: Model Results

| Model       | $R^2$ | Hit Rate |
|-------------|-------|----------|
| VARX (Base) | 0.89  | 49.44%   |
| VARMAX-SVR  | 0.99  | 49.20%   |
| VARMAX-GBR  | 0.97  | 48.92%   |
| VARMAX-LGBM | 0.90  | 49.44%   |
| VARMAX-LSTM | 0.88  | 49.01%   |
| VARMAX-MLP  | 0.98  | 50.81%   |

Backtesting is done with initial capital of 10,000. Figure 6.1, shows the equity curves over time, overlayed with the S&P500 simple buy-and-hold over the same period of January 2024 to January 2026. In the top diagram in every graph, the **Orange** line represents S&P500 buy-and-hold equity curve and the **Green** line represents the model's equity curve. Respectively the lower diagram represents the drawdown over time.

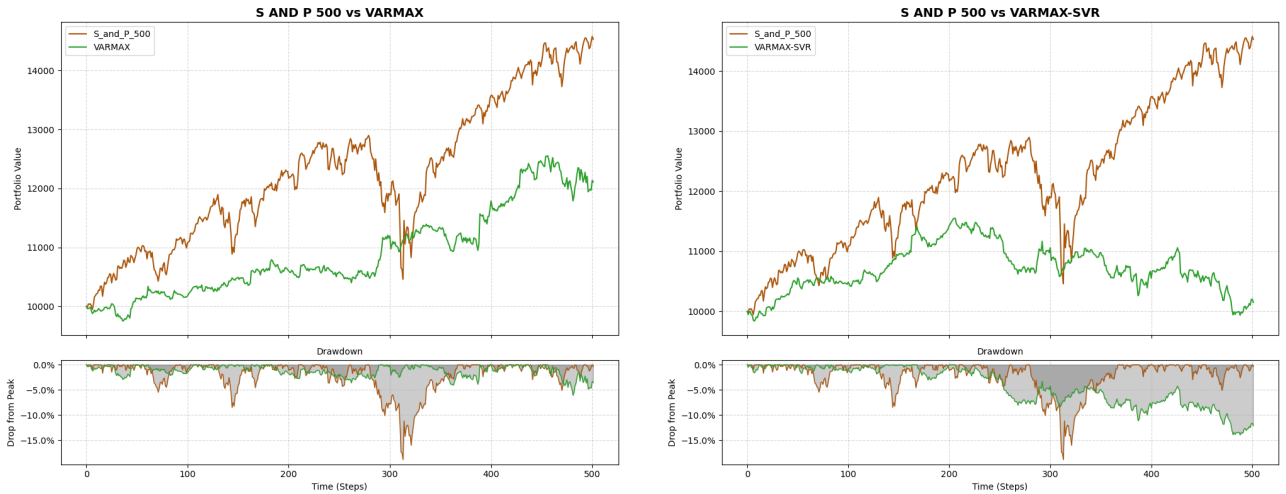


Figure 8: Backtesting Results

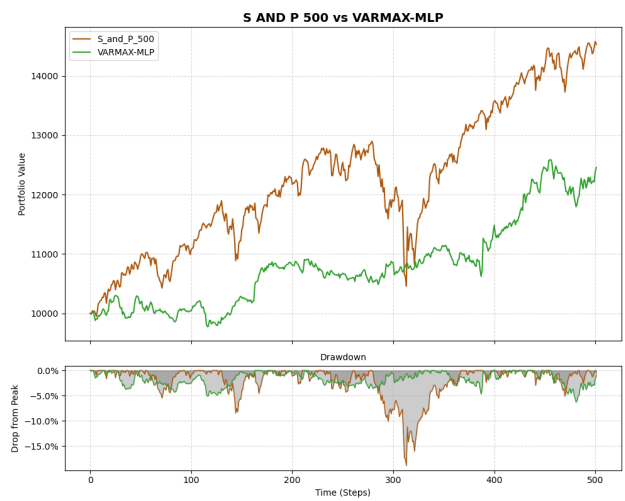
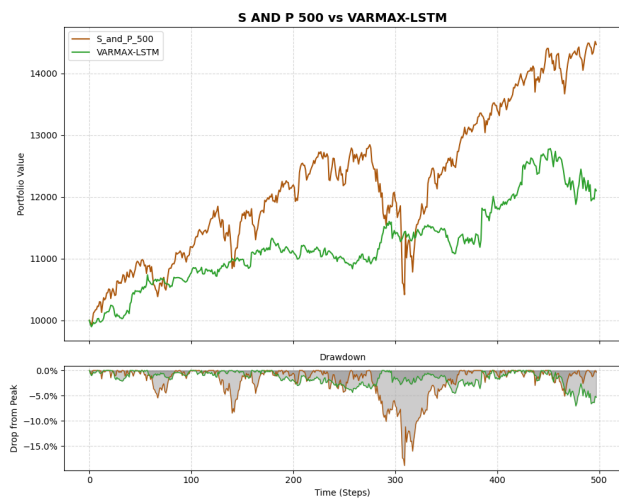
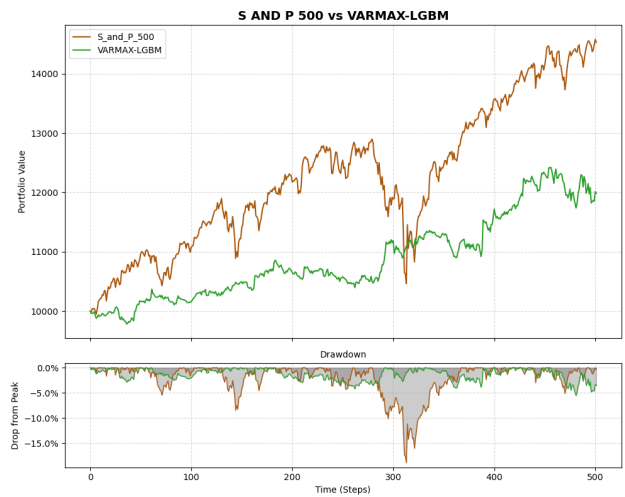
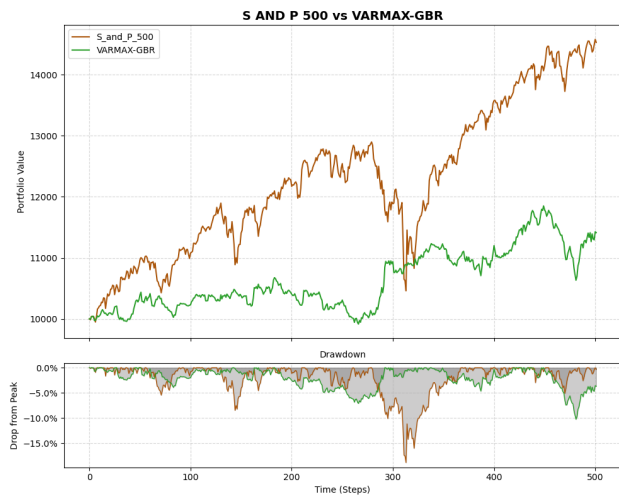


Figure 9: Backtesting Results

## 6.2 Backtest Financial Metrics

We compute the following financial metrics on  $C$ :

### 1. Annualized Sharpe Ratio:

$$\text{Sharpe} = \frac{(\sum_{i=2}^D \frac{C_i - C_{i-1}}{C_{i-1}} / (D-1))}{\sqrt{\sum_{i=2}^D \frac{(\frac{C_i - C_{i-1}}{C_{i-1}} - (\sum_{i=2}^D \frac{C_i - C_{i-1}}{C_{i-1}} / (D-1)))^2}{(D-1)}}} \cdot \sqrt{252} \quad (18)$$

### 2. Annualized Sortino Ratio:

$$\text{Sortino} = \frac{(\sum_{i=2}^D \frac{C_i - C_{i-1}}{C_{i-1}} / (D-1))}{\sqrt{\sum_{i=2}^D \frac{\min(0, \frac{C_i - C_{i-1}}{C_{i-1}} - (\sum_{i=2}^D \frac{C_i - C_{i-1}}{C_{i-1}} / (D-1)))^2}{(D-1)}}} \cdot \sqrt{252} \quad (19)$$

### 3. Compound Annual Growth Rate:

$$\text{CAGR} = \left( \frac{C_D}{C_1} \right)^{\frac{252}{D}} - 1 \quad (20)$$

### 4. Maximum Drawdown:

$$\text{Maximum Drawdown} = \max_{t \in \{1, \dots, D\}} \left( \frac{\max_{s \in \{1, \dots, t\}} C_s - C_t}{\max_{s \in \{1, \dots, t\}} C_s} \right) \quad (21)$$

### 5. Calmar Ratio:

$$\text{Calmar} = \frac{\text{CAGR}}{\text{Maximum Drawdown}} \quad (22)$$

Table 2 summarizes the financial metrics of the strategies. Financial metrics are the most appropriate way to benchmark these strategies against each other, comparing how they perform at a backtest. Table 3 shows the daily turnover of the strategies and the net CAGR. This presents us with more realistic expected returns, since in the real world a small amount of transaction costs are deducted with every trade.

Table 2: Portfolio Performance and Risk Metrics (Out-of-Sample)

| Model       | Sharpe | Sortino | CAGR     | Max Drawdown         | Calmar |
|-------------|--------|---------|----------|----------------------|--------|
| S&P 500     | 1.25   | 1.58    | 20.54%   | 18.90%               | 1.08   |
| VARX (Base) | 1.14** | 1.89**  | 10.07%** | 6.10% <sup>†</sup>   | 1.40** |
| VARMAX-SVR  | 0.10   | 0.14    | 0.5%     | 14.17%               | 0.03   |
| VARMAX-GBR  | 0.83   | 1.36    | 6.8%     | 6.65%                | 1.03   |
| VARMAX-LGBM | 1.07** | 1.79**  | 9.51%**  | 5.53% <sup>†</sup>   | 1.72** |
| VARMAX-LSTM | 1.12** | 1.81**  | 10.12%** | 7.07% <sup>†</sup>   | 1.43** |
| VARMAX-MLP  | 1.29** | 2.21**  | 11.86%** | 6.57%** <sup>†</sup> | 1.80** |

Note: \* =  $p < 0.1$ , \*\* =  $p < 0.05$  against Random Signals. <sup>†</sup> =  $p < 0.1$ , <sup>††</sup> =  $p < 0.05$  against S&P 500.

Table 3: Turnover

| Model       | Turnover (Daily) | Net CAGR* |
|-------------|------------------|-----------|
| VARX (Base) | 5.58%            | 8.67%     |
| VARMAX-SVR  | 35.28%           | -8.39%    |
| VARMAX-GBR  | 16.48%           | 2.65%     |
| VARMAX-LGBM | 4.95%            | 8.26%     |
| VARMAX-LSTM | 5.37%            | 8.77%     |
| VARMAX-MLP  | 13.98%           | 8.34%     |

\* Theoretical adjusted returns using transaction cost of 0.10% per trade.

### 6.3 Some Additional Checks

Table 4 shows the results of some additional checks.

- Pearson Correlation Coefficient of Model returns and S&P 500 returns. To test if returns are linearly uncorrelated with S&P 500
- Skew of returns. A high positive skew indicates more frequent but small losses and less frequent but large gains.
- Truncated CAGR: CAGR excluding the top 1 percentile of returns. a very low value compared to CAGR indicates that returns were driven mostly by a few outlier events.
- Positive Sortino: Proportion of days where Sortino stayed positive (we calculate 2-month rolling sortino). A high percentage signals that the strategy stays consistent throughout the testing period.
- Beta: Measures volatility against a benchmark(S&P500)

Table 4: Additional Checks

| Model         | Correlation Coefficient | Skew Of returns | Truncated CAGR | Positive Sortino | beta  |
|---------------|-------------------------|-----------------|----------------|------------------|-------|
| S&P 500       | 1                       | 0.71            | 8.62%          | 74%              | 1     |
| VARMAX (Base) | 0.0039                  | 1.30            | 3.69%          | 64%              | 0.002 |
| VARMAX-SVR    | -0.038                  | -0.35           | -3.18%         | 51%              | -0.02 |
| VARMAX-GBR    | -0.032                  | 0.40            | 1.61%          | 56%              | -0.01 |
| VARMAX-LGBM   | 0.0038                  | 1.27            | 3.15%          | 63%              | 0.002 |
| VARMAX-LSTM   | 0.029                   | 1.11            | 3.88%          | 65%              | 0.018 |
| VARMAX-MLP    | 0.0099                  | 1.56            | 4.89%          | 64%              | 0.005 |

## 7 Conclusion and suggestions for improvement

### 7.1 What these strategies do better than simple buy and hold

In this study, I showed that these strategies showed a much lower drawdown rate, and higher sortino compared to S&P500 buy and hold. These strategies are best suited for a leveraged portfolio where the low volatility of my strategy and a higher sortino (Table 2) helps in generating consistent alpha. The correlation checks summarized in Table 4 show that the returns of these strategies are uncorrelated with S&P500. In terms of Net CAGR, VARMAX-LSTM and VARMAX-MLP were the clear winners. VARMAX-MLP had the highest calmar ratio, thus the highest return per risk taken.

### 7.2 Which Strategy/ies improved the linear model ?

VARMAX-MLP is the best case where the hybrid model improved the linear model. It beat the base linear model on every financial metric we tested it on (Table 2).

### 7.3 Limitations of this study and future research

The core limitations of this study were computational and data limitations. We had to restrict our ML parameter space to a small grid which was selected to not overfit on the data. Single Neural Network layers and shallow tree depths were given. The problem can be solved by expanding the grid of parameters, so that we may find better parameters. We used a simple set of features to understand how the models will compare against each other and the base model. Future research should engineer more complex or varied features to find patterns. The data limitation is the fact that we used 8 years of data from 2016 to 2024 for training and 500 days for model testing and backtesting the strategy. Future research can expand the training set and the testing period to get an even more generalized view and a stress test of the strategies under different market conditions or regimes. We have only used closing price data as our primary variable. We can extract more patterns by using other forms of daily data, such as opening price, market cap, volume, etc. I expect the Neural Networks to have the most potential in finding complex patterns in the data.